

QUANTUM CLUB

The 7 Algorithms Every Quant **Must Master**

An intuitive guide to the models that move the markets, from Black-Scholes to machine learning, explained without getting lost in the mathematics.

FREE GUIDE . INTRODUCTORY LEVEL

Black-Scholes . Monte Carlo . Markowitz . Pairs trading . Kalman filter . GARCH . Machine learning

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One page per algorithm: the idea in plain language, what actually drives it, where it breaks, and a chart of what it looks like. No derivations. If you want those, that is what the program is for.

Black-Scholes

The fair price of an option

Before 1973 nobody could say what an option was worth. Black, Scholes and Merton answered it with an argument that still feels like a magic trick: build a portfolio of the stock and the option that cancels its own risk, and it must earn the risk-free rate, or someone gets free money.

- **The intuition**

You are not forecasting the stock. You are pricing the cost of hedging it until expiry.

- **What drives the price**

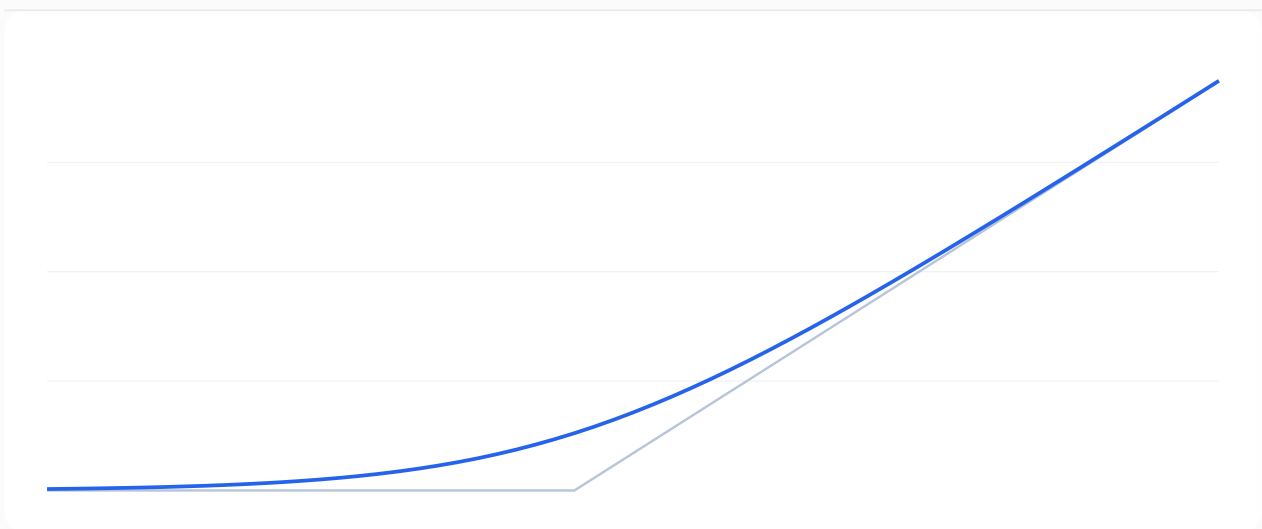
Time to expiry and volatility. More of either means more uncertainty, and uncertainty is what an option sells.

- **Where it breaks**

It assumes constant volatility and no jumps. Markets have both, which is why the volatility smile exists.

$$C = S N(d_1) - K e^{(-rT)} N(d_2)$$

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Blue: what the option is worth today. Grey: what it pays at expiry.

Monte Carlo

When there is no formula, simulate

Most real instruments have no closed-form price. Monte Carlo sidesteps the algebra entirely: simulate thousands of possible futures for the underlying, compute the payoff in each one, and average them back to today.

- **The intuition**

An expected value is just an average. If you cannot integrate it, sample it.

- **The cost**

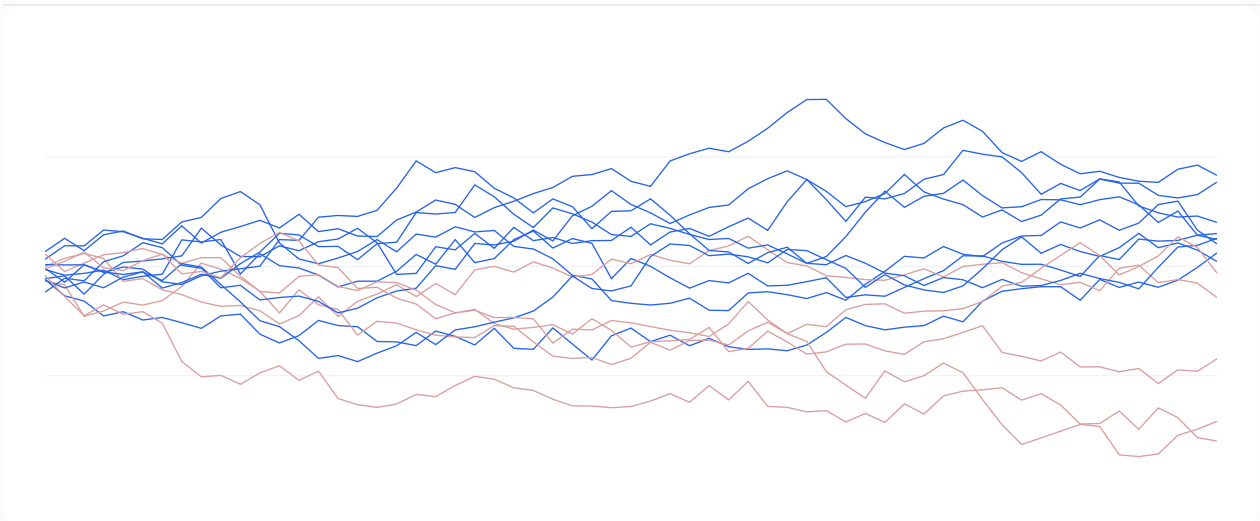
Accuracy improves with the square root of the number of paths. Four times the paths buys you twice the precision.

- **Why it wins**

It handles path dependency, multiple assets and exotic payoffs that no formula covers.

```
Price =  $e^{(-rT)}$  x average of payoff(path_i)
```

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Fourteen of ten thousand simulated paths. The price is their average.

Markowitz

Turning risk into a decision

Markowitz showed that a portfolio is not the sum of its holdings. What matters is how they move together: two risky assets that fall at different times combine into something calmer than either one alone.

- **The intuition**

Diversification is not owning more things. It is owning things driven by different forces.

- **The efficient frontier**

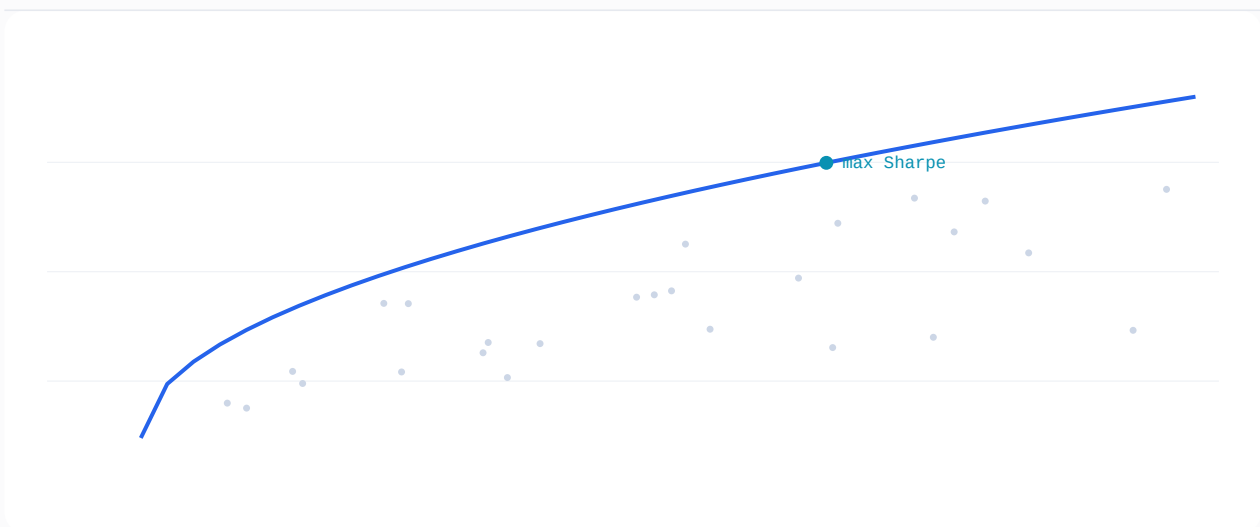
For every level of risk there is one allocation with the highest expected return. That curve is the frontier.

- **The catch**

It is exquisitely sensitive to the expected returns you feed it, and those are the hardest thing to estimate.

minimise $w' S w$ subject to $w' \mu = \text{target}$

AT A GLANCE



Each dot is a portfolio. The curve is the best you can do.

Pairs trading

Betting on the gap, not the level

Two related assets drift apart, then snap back. Statistical arbitrage takes no view on where the market is going, only that the distance between two things is temporarily wider than it should be.

- **The intuition**

You are trading a spread, not a direction. If both legs fall, the position can still win.

- **Ornstein-Uhlenbeck**

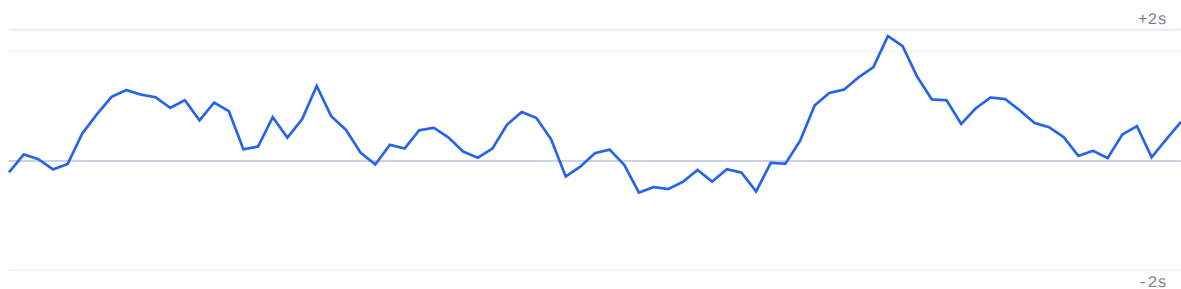
The mathematics of a spring: the further the spread stretches, the harder it is pulled back.

- **The risk**

The relationship can simply end. A spread that stops mean-reverting is no longer a trade, it is a loss.

$$d(\text{spread}) = \theta (\mu - \text{spread}) dt + \sigma dW$$

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You trade the gap between two assets, not their direction.

Kalman filter

The signal under the noise

Every price you observe is the truth plus noise. The Kalman filter estimates the hidden state you actually care about, updating its belief each time new data arrives and weighting it by how much it trusts the measurement.

- **The intuition**

It is a running compromise between what you predicted and what you just saw.

- **Why quants use it**

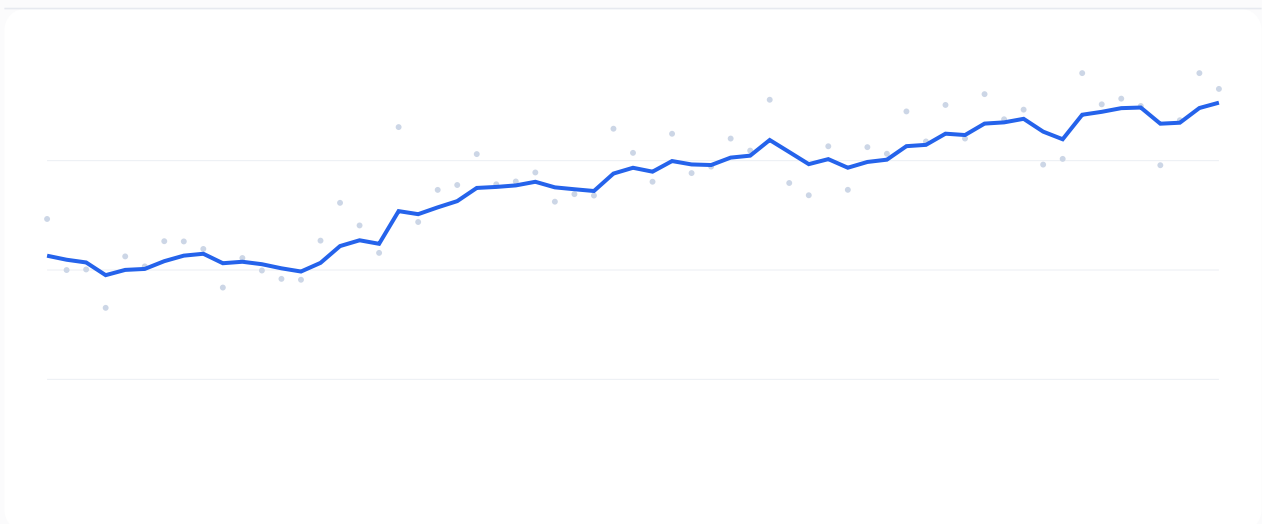
Hedge ratios drift. The filter re-estimates them continuously instead of refitting a window by hand.

- **The nice part**

It is recursive: it needs only the last estimate, not the whole history.

```
estimate = prediction + K x (observation - prediction)
```

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Dots are what you observe. The line is what you believe.

GARCH

Volatility arrives in bursts

Volatility is not constant and it is not random noise either. Quiet weeks follow quiet weeks and turmoil follows turmoil. GARCH models that clustering directly, which is why it remains the workhorse of risk forecasting.

- **The intuition**

Today's volatility depends on yesterday's volatility and yesterday's surprise.

- **What it gives you**

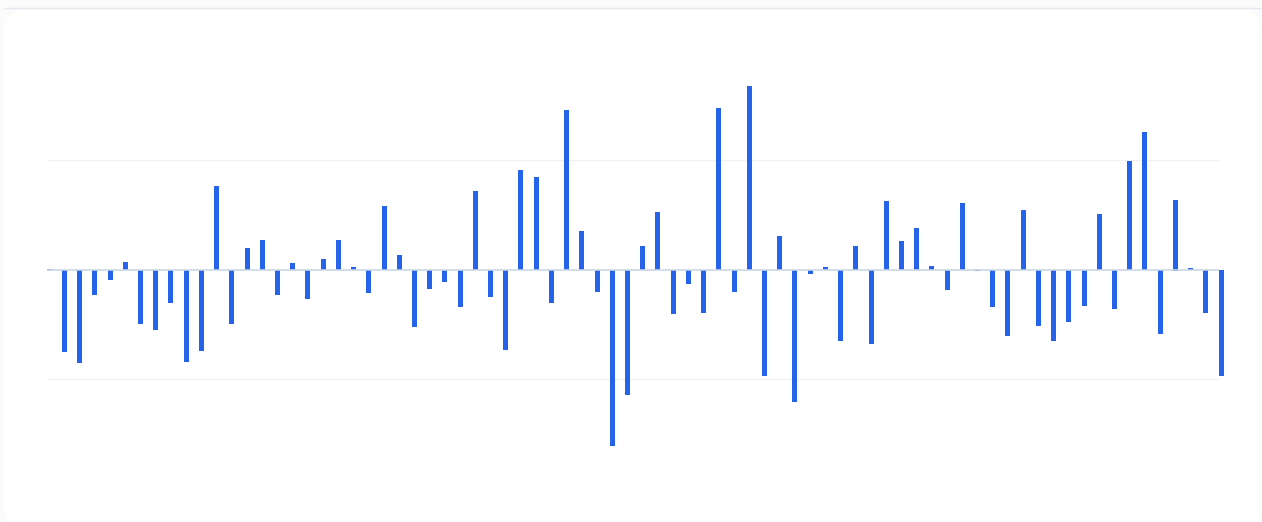
Forecast volatility, which feeds position sizing, VaR and option pricing.

- **Its limit**

It reacts to shocks, it does not anticipate them.

$$\sigma^2_t = \omega + \alpha e^2_{t-1} + \beta \sigma^2_{t-1}$$

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Calm follows calm; storms cluster. That is what GARCH captures.

Machine learning

Where it helps, and where it does not

Markets are the hardest setting there is for machine learning: the signal is faint, the data is limited, and the rules change while you are learning them. That does not make it useless. It makes it a tool with a narrow, real job.

- **Where it works**

Feature-rich, repeatable problems: execution, classification, filling gaps in messy data.

- **Where it fails**

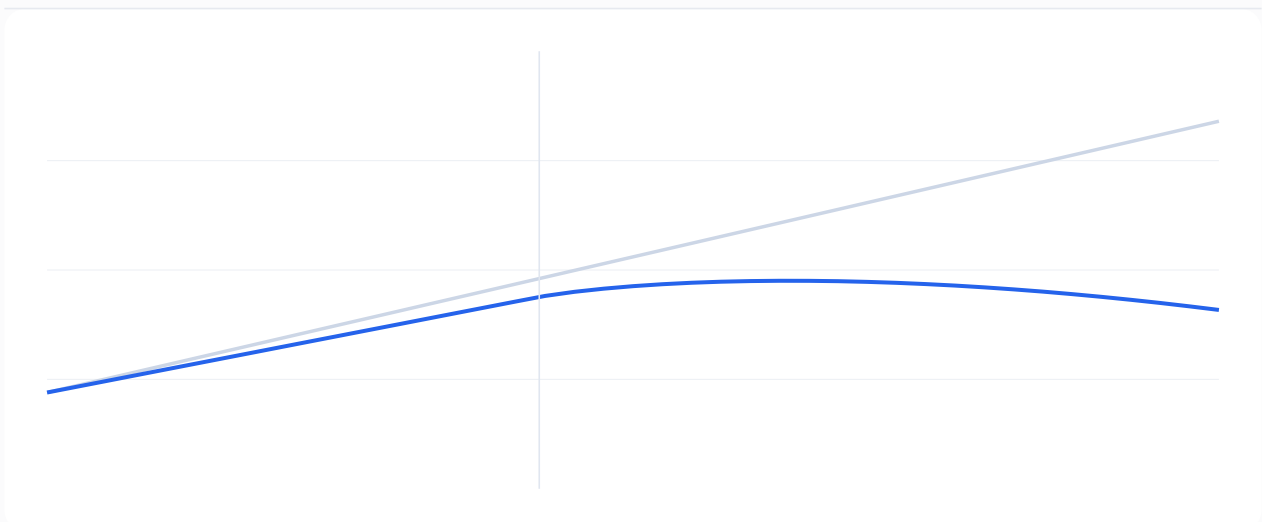
Naive price prediction. Enough parameters will fit any past, and that past does not repeat.

- **The discipline**

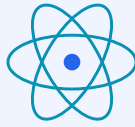
Out-of-sample testing and honesty about how many ideas you tried before this one worked.

The model is the easy part. The validation is the job.

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In-sample keeps improving. Out-of-sample is where the truth is.



Where this goes next

You have the map. What it does not give you is the thing that actually gets you hired or gets a strategy live: writing these models yourself, in code, and knowing why each line is there.

- Six weeks, from zero to a working pricing engine or backtest.
- Every Jupyter notebook, yours to keep and run.
- A private Discord where questions get answered.
- Lifetime access, including everything added later.

Quantum Club

From zero to quant finance in six weeks.

[See the program ->](#)

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